

```
def train_gates(model, patterns, epochs=100):
    optimizer =
torch.optim.Adam(model.parameters(),
lr=0.001)

    for epoch in range(epochs):
        total_loss = 0

        # Генерация потока спайков
        spikes =
generate_spike_stream(patterns)

        result = model(spikes)
        xin = result['gated_input']
        h1 = result['primary_memory']

        # Loss 1: Ворота должны
восстанавливать паттерны
        recon_loss =
F.mse_loss(xin.mean(dim=0),
patterns.mean(dim=0))
```

```
# Loss 2: Хебб между xin и h1  
(импликативная связь)
```

```
hebb_loss = -(xin *  
h1.unsqueeze(1)).mean()
```

```
loss = recon_loss + 0.1 * hebb_loss
```

```
optimizer.zero_grad()
```

```
loss.backward()
```

```
optimizer.step()
```

```
if epoch % 20 == 0:
```

```
    print(f"Epoch {epoch}:
```

```
    Loss={loss.item():.4f}, "
```

```
        f"Sync
```

```
    quality={result['knowing_strength']:.3f}")
```